Exp : 2 Study of different types of network cable

Aim:

Study of different types of network cables and understand the procedure for making ethernet straight - through and crossover cables.

Theory:

Network cables:

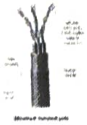
Cable type	category	Maximum data transmission	Pros/ Cons	Use
UTP	category 3	10 bps	Pros: • cheaper in cost • easy to install as they have a smaller diameter	10 Base-T ethernet
	category 5	upto 100 mbps		fast ethernet
	category 5e	1 Gbps		gigabit ethernet
			Cons: More prone to (EMI)	fast ethernet gigabit ethernet
STP	category 6, 6a	10 Gbps	Pros: • Shielded • Faster than UTP.	gigabit ethernet 10 G (55m) Widely used in data centre.
SSTP	category 7	10 Gbps	Cons: • Expensive • Greater installation effort	gigabit ethernet 10G ethernet (100m)

Network cables:

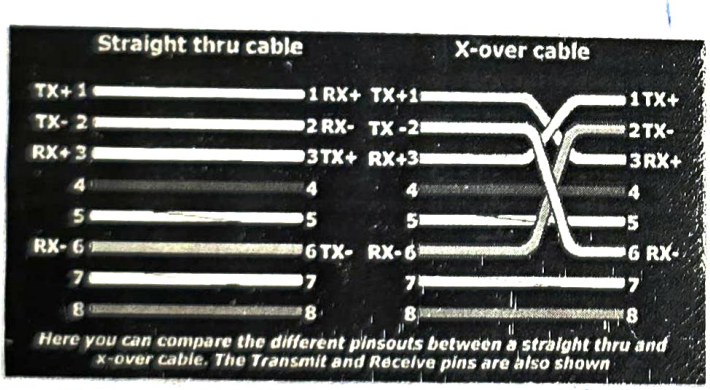




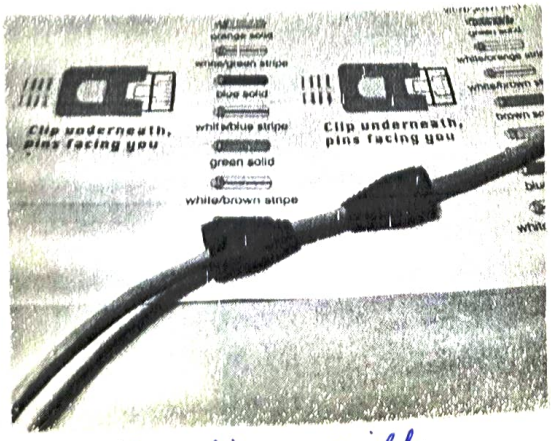








Different between cross cable and straigh cable



threading shields into cables

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss Cons: • Cost • Size is bulky	Speed of singnal is 500m Television network high speed internet
fibre optics cable	single mode multi mode	100 Gbps	Pros: • High speed • High band width Cons: • Expensive • Requires skilled installers.	Maximum distance of fibre optics cable is around 100 metres

6) Make your own ethernet cross-over cable.

- Tools and parts required:
- Ethernet cabling. CAT5e is certified for gigabait support, but CAT5 cabling works as well,
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cable.
- Two RJ45 plugs.
- Optional two plug shields.

Procedure:

1) Insert the cable boot.

2) Strip about 1.5 cm of the outer insulation

3) Untwist the four wire pairs.

4) Arrange the wires according to the required colour code.

5) Trim the wires evenly

6) Insert the wires fully into the RJ 45 connector

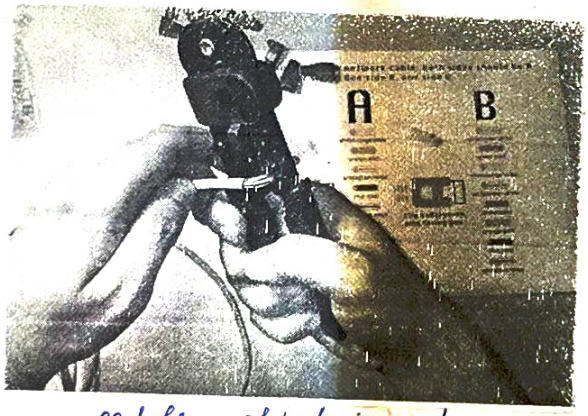
7) Crimp the connector using the crimping tool.

8) Repeat the same process for the other end.

9) Test the cable using a cable-tester or by connecting two devices.

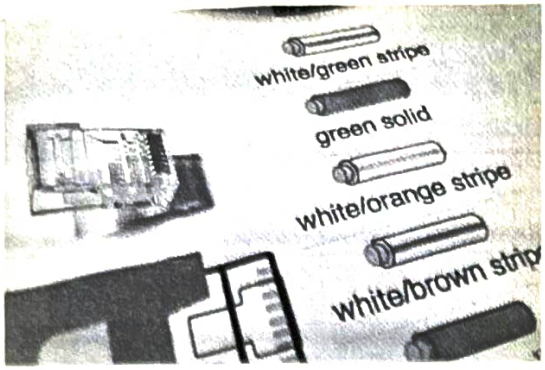


untangle the wires to form four-twisted pairs.

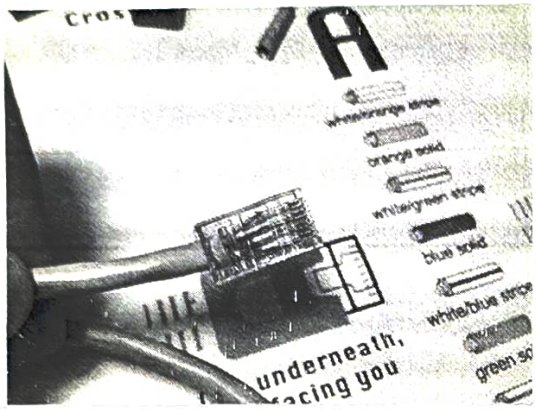


cable shielding from both ends

Network cables



- Arrange the wires evenly in order, trim any uneven ends, insert them into RJ45 plug with the clip facing away and gold pins toward you



- The wires in the plug are inserted into the crimping tool and pushed down.

Result:

Study of different types of network cables through cable and crossover is done successfully.